Carlos Damian Rojas

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Education

The University of Texas at Austin December 2027
Bachelor of Science in Electrical and Computer Engineering, Minor in Robotics *Austin, TX*
Relevant Coursework: Operating Systems, Computer Architecture, Embedded Systems Design Lab, Digital Logic Design, Data Structures & Algorithms, Software Testing, Circuit Theory, Linear Algebra

Experience

Amazon May 2026 – Aug 2026
Software Development Engineer Intern *Seattle, WA*
- Designed and shipped an **end-to-end personalization system** for the Photos team generating customized video content
- Built an occasion-agnostic **SQS event handler** on AWS Lambda dispatching to per-occasion recipes and writing S3 artifacts
- Architected a fail-closed **allowlist gating model** ensuring safe rollout and controlled testing across multiple **AWS accounts**
- Owned the full technical stack across **8 Python and Java packages** managing recipe logic, infrastructure, and persistence
- Shipped **2 occasions end-to-end** to production behind a weblab gate, merging **19 code reviews** across 8 packages

Texas Information Security Office March 2025 – August 2025
Software Engineering Intern, Cybersecurity *Austin, TX*
- Engineered **Python CLI tools** and Linux **automation scripts** to analyze logs, cutting investigation prep time by 40%
- Performed **10+** simulated **security triages** by analyzing Splunk SIEM logs, packet captures in Wireshark, and HTTP requests
- Built and executed **workflows** for web-app vulnerability **analysis**, validating issues like SQL injection and parameter tampering
- Enhanced security telemetry pipelines by **collaborating** with cybersecurity analysts and infrastructure engineers

UTSA Unmanned Systems Lab January 2024 – August 2024
Machine Learning Research Intern *San Antonio, TX*
- Benchmarked an event-triggered **PPO** variant against standard PPO on **MuJoCo** Half-Cheetah in **PyTorch**
- Ran **90+ training configurations** from Bash, grid-searching learning rate, rollout length, batch size, and clip range
- Produced the Half-Cheetah figures for an **ACC 2025** paper: within ~10% of PPO's reward, ~**20% fewer** control updates

Projects

PintOS Operating System Kernel | *C, x86*
- Implemented **11 Unix system calls** in a 32-bit x86 kernel with safe user memory validation
- Designed **semaphore-based** parent-child synchronization using heap-allocated child-info structs, correctly handling race orderings
- Built **clock-algorithm page eviction** with a swap table, pinning frames during file I/O syscalls
- Rebuilt inodes as **indexed blocks** (12 direct, indirect, doubly indirect) with file growth and per-inode locks

FreeRTOS CAN ECU Network on TI MSPM0 | *C, FreeRTOS, CAN, TI MSPM0, Python*
- Built FreeRTOS firmware for a 2-ECU **CAN** network on **TI MSPM0** (Cortex-M0+) with fault and **limp-home** state machines
- Wrote MCAN (500 kbit/s), ADC, I^2C, SPI and PWM drivers on TI DriverLib; CAN messages generated from a **.dbc** file
- Verified the firmware logic with 92 **Unity** tests, incl. power-loss injection at every flash write

FleetFlight: Battery Firmware Model Checker | *Python, React, TypeScript, three.js* Base Power Hackathon
- Built a **bounded model checker** for distributed battery firmware, searching fault orderings across BMS, hub, and inverter
- Explored **1.29M states in 38 s** with BFS, returning the **shortest failing sequence** for each violated safety invariant
- Replayed counterexamples through the same transition code with **100/100 identical trace hashes** across runs

Custom Guitar Hero Controller | *C, USB HID, UART, ADC*
- Wrote C firmware for the **Raspberry Pi Pico** to scan mechanical switches with interrupt **debouncing** with sub 5 ms latency
- Debugged electrical issues using **multimeters**, **logic analyzers**, and serial monitoring tools to ensure deterministic timing
- Designed and 3D-printed a fully custom Guitar Hero controller with a fretboard **PCB** and **USB encoder harness**

Technical Skills

Languages: C, Python, bash, Assembly, Verilog, Java
Technologies: Linux/Unix, FreeRTOS, CAN, pytest, GitHub Actions, Logic Analyzer, Oscilloscope, UART, KiCad
AI/ML Technologies: PyTorch, NumPy, Matplotlib, Claude Code

Leadership / Extracurricular

Texas Club Rugby Team Fall 2024 – Present
Team Captain, Vice President
- Led and represented **30+ teammates**, coordinating two weekly practices and travel logistics for **10+ away games**
- Captained the team to **3 in-state tournament** victories and qualification for the **national tournament** in 2025

Association of Latino Professionals for America (ALPFA) Spring 2026 – Present
Technology & Innovation Advisor, Professional Development Committee
- Building a **web app for live interview scoring** at the chapter's speed-interviewing capstone, replacing manual tallying
- Ran attendance tracking and **anonymous feedback forms** across **6 events** of **20+ members** to guide topic and format changes